

3.2 Linear Momentum & Conservation

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.2 Linear Momentum & Conservation
Difficulty	Medium

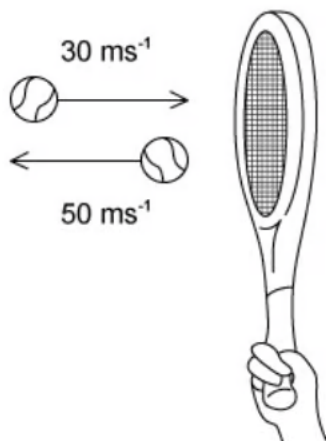
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A tennis ball of mass 60 g is struck by a tennis racket. The velocity of the ball changes as shown



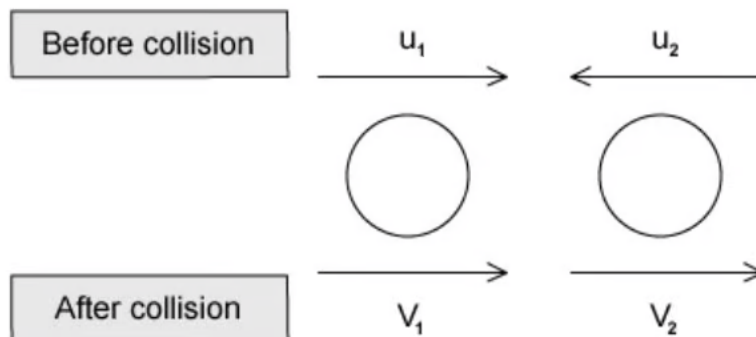
What is the magnitude of the change in momentum of the ball?

- A. $1.2 \times 10^3 \text{ kg m s}^{-1}$
- B. $4.8 \times 10^3 \text{ kg m s}^{-1}$
- C. 1.2 kg m s^{-1}
- D. 4.8 kg m s^{-1}

[1 mark]

Question 2

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision, and v_1 and v_2 after collision, in the directions shown below



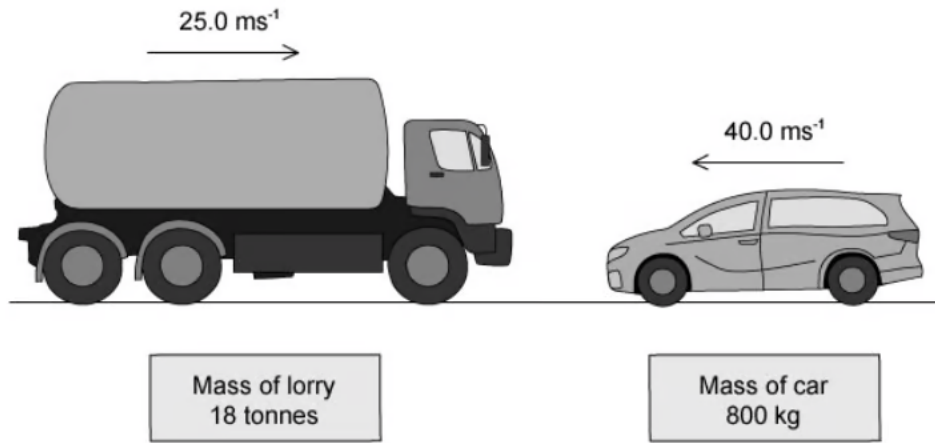
Which equation is correct if the collision is perfectly elastic?

- A. $u_1 - u_2 = v_2 + v_1$
- B. $u_1 - u_2 = v_2 - v_1$
- C. $u_1 + u_2 = v_2 - v_1$
- D. $u_1 + u_2 = v_2 + v_1$

[1 mark]

Question 3

A lorry of mass 18 tonnes is travelling at 25.0 m s^{-1}



A car of mass 800 kg is travelling at 40.0 m s^{-1} towards the lorry.

What is the magnitude of the total momentum?

- A. 282 kN s
- B. 418 kN s
- C. 482 kN s
- D. 1222 kN s

[1 mark]

Question 4

The diagram shows the masses and velocities of two trolleys about to collide



After the impact they move off together

What is the total kinetic energy of the trolleys after the collision?

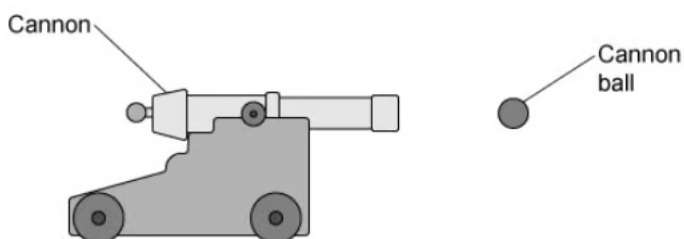
- A. 6.0 J
- B. 3.7 J
- C. 0.35 J
- D. 0.02 J

[1 mark]

Question 5

The diagram shows a cannon ball fired from a cannon

The mass of the cannon is 1600 kg and the mass of the cannon ball is 20 kg



The recoil velocity of the cannon is 5 m s^{-1} horizontally

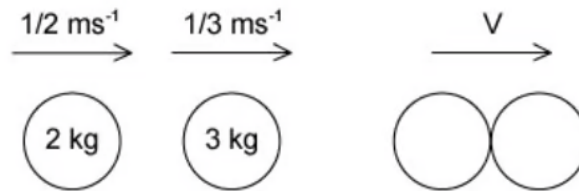
What is the horizontal velocity of the cannon ball?

- A. 100 m s^{-1}
- B. 320 m s^{-1}
- C. 400 m s^{-1}
- D. 8000 m s^{-1}

[1 mark]

Question 6

A ball of mass 2 kg travelling at $\frac{1}{2}\text{ m s}^{-1}$ strikes a ball of mass 3 kg travelling at $\frac{1}{3}\text{ m s}^{-1}$. Both balls are moving along the same straight line as shown



After collision, both balls move at the same velocity v

What is the magnitude of the velocity v ?

- A. $\frac{1}{5}\text{ m s}^{-1}$
- B. $\frac{2}{5}\text{ m s}^{-1}$
- C. $\frac{3}{2}\text{ m s}^{-1}$
- D. $\frac{5}{2}\text{ m s}^{-1}$

[1 mark]

Question 7

A molecule of mass m travelling at speed v hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the momentum and in the kinetic energy of the molecule caused by the collision?

	Change in momentum	Change in kinetic energy
A	0	0
B	mv	mv^2
C	mv^2	0
D	$2mv$	0

[1 mark]

Question 8

A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

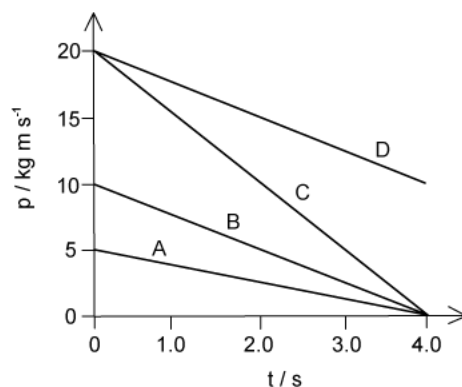
	Total momentum before collision / kg m s^{-1}	Total momentum after collision / kg m s^{-1}	Total kinetic energy before collision / J	Total kinetic energy after collision / J
A	6	6	90	30
B	6	6	30	90
C	6	2	90	30
D	6	6	90	90

[1 mark]

Question 9

A resultant force of 5 N acts on a body for a time of 4.0 s

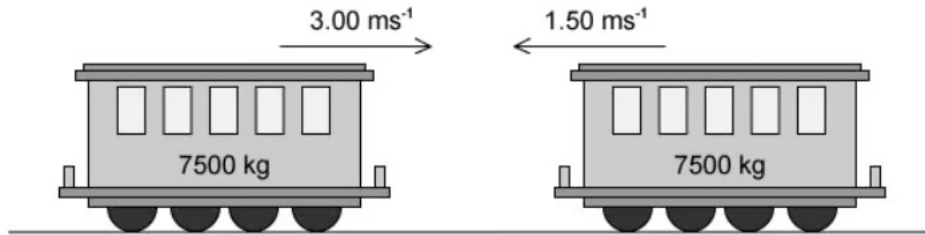
Which graph could show the variation with time t of the momentum p of the body?



[1 mark]

Question 10

Two train carriages each of mass 7500 kg roll toward one another on a level track. One is travelling at 3.00 m s^{-1} and the other at 1.50 m s^{-1} as shown.



They collide and join together.

What is the kinetic energy lost during the collision?

- A. 4200 J
- B. 7500 J
- C. 11250 J
- D. 38000 J

[1 mark]

Model Answers: Medium

1

The correct answer is **D** because:

- The ball changes direction after it is hit by the racket, so we will have to set a positive and a negative direction in order to calculate the correct value
- Let's set the positive direction in the direction of the ball after the collision (in the 50 m s^{-1} direction)
- Change in momentum is equal to
 - $\Delta p = m \Delta v = m(\text{final } V - \text{initial } V)$
 - $\Delta p = (60 \times 10^{-3})(50 - (-30))$
 - $\Delta p = 4.8 \text{ kg m s}^{-1}$

A is incorrect because this is the answer you get if you don't convert the mass from g to kg and if you don't set a positive and negative direction and add the velocities together

B is correct because this is the answer you get if you don't convert the mass from g to kg

C is incorrect because this is the answer you get if you don't set a positive and negative direction and add the velocities together

2

The correct answer is **C** because:

- In a perfectly elastic collision, **both** momentum and kinetic energy are conserved
- In an elastic collision: the relative speed of approach before the collision must be equal to the relative speed of separation after the collision
- Before the collision: the two spheres are moving towards each other, they are approaching each other
 - The relative speed of approach is $u_1 + u_2$
- After the collision: the spheres move in the same direction, the first sphere is

moving towards the second sphere and the second sphere is moving away from the first – this is an important distinction to make as it tells us v_2 must be greater than v_1

- The relative speed of separation is $v_2 - v_1$
- So, the relative speeds before and after the collision are equal to
 - $u_1 + u_2 = v_2 - v_1$

3

The correct answer is **B** because:

- Momentum is a vector quantity, so we need to set out a positive and negative direction
- Let's define the positive direction as the direction towards the right
- Momentum is equal to $p = mv$
- Total momentum = Momentum of lorry + Momentum of car
- Total momentum = $(18000 \times 25) + (800 \times -40)$

$$= 418000 \text{ N s}$$

$$= \mathbf{418 \text{ kN s}}$$

4

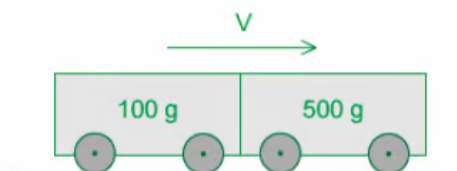
The correct answer is **C** because:

- After the collision, the trolleys move together as a single trolley of mass 600 g and velocity v
- We can find the value of v using the principle of conservation of momentum:
 - The sum of momentum before the collision equals the sum of momentum after the collision
- Momentum before: $(100 \times 4) + (500 \times 0.5) = 400 + 250 = 650$
 - Note: this is the same as writing

$$(100 \times 10^{-3} \times 4) + (500 \times 10^{-3} \times 0.5) = 650 \times 10^{-3}$$

- Momentum after: $(100 + 500)v = 600v$
- Therefore, $650 = 600 v$

- $v = \frac{650}{600} = 1.08 \text{ m s}^{-1}$
- Kinetic energy $E_k = \frac{1}{2}mv^2$
 - $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times (600 \times 10^{-3}) \times 1.08^2 = \mathbf{0.35J}$



5

The correct answer is **C** because:

- Before the cannon ball is fired both the cannon and the ball are stationary so the initial momentum is zero
- The cannon recoils in the opposite direction to the direction of motion of the cannon ball, so we must define one of the directions to be positive and one to be negative
- By the principle of conservation of momentum, the sum of momentum before the collision equals the sum of momentum after the collision
 - $0 = (1600 \times 5) - 20 v$
 - $20 v = 8000$
 - $v = 400 \text{ m s}^{-1}$

6

The correct answer is **B** because:

- After the collision, the balls move together as a single object of mass 5 kg and velocity v
- We can find the value of v using the principle of conservation of momentum:
- The sum of the momentum before the collision equals the sum of the momentum after the collision
 - $\left(2 \times \frac{1}{2}\right) + \left(3 \times \frac{1}{4}\right) = 5 v$

- $5v = 2$
- $v = \frac{2}{5} m s^{-1}$

7

The correct answer is **D** because:

- For any collision in a closed system, momentum is always conserved
- Before the collision:
 - Momentum of the molecule = mv
- After the collision:
 - The molecule will still have the same momentum but it will move in the opposite direction so the momentum of the molecule = $-mv$
- Change in momentum = $mv - (-mv) = 2mv$
- The collision is elastic so the kinetic energy is conserved, therefore, the change in kinetic energy = **0**

A is incorrect because the change in momentum cannot be zero because the molecule changes direction

B is incorrect because the change in kinetic energy must be zero because the collision is elastic meaning the kinetic energy is conserved and does not change

C is incorrect because mv^2 has the units of energy

8

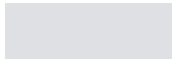
The correct answer is **A** because:

- For any collision in a closed system, momentum is always conserved, so the total momentum before and after the collision should be the same
- The collision is inelastic, meaning that kinetic energy is not conserved, so the kinetic energy after the collision should be less than the energy before

B is incorrect because the kinetic energy before the collision cannot be lower than the kinetic energy after the collision

C is incorrect because momentum is always conserved, so the momentum should be the same before and after the collision

D is incorrect because the collision is inelastic, meaning the energy is not conserved, so the energy should be less than the initial energy



kinetic energy = 0

A is incorrect because the change in momentum cannot be zero because the molecule changes direction

B is incorrect because the change in kinetic energy must be zero because the collision is elastic meaning the kinetic energy is conserved and does not change

C is incorrect because mv^2 has the units of energy

8

The correct answer is **A** because:

- For any collision in a closed system, momentum is always conserved, so the total momentum before and after the collision should be the same
- The collision is inelastic, meaning that kinetic energy is not conserved, so the kinetic energy after the collision should be less than the energy before

B is incorrect because the kinetic energy before the collision cannot be lower than the kinetic energy after the collision

C is incorrect because momentum is always conserved, so the momentum should be the same before and after the collision

D is incorrect because the collision is inelastic, meaning the energy is not conserved, so the energy should be less than the initial energy

9

The correct answer is **C** because:

- Force is equal to the rate of change of momentum
- In other words, force is equal to the gradient $\left(\frac{\Delta y}{\Delta x}\right)$ of a momentum-time graph

$$\circ F = \frac{\Delta p}{\Delta t}$$

- The gradient of the line should be equal to 5 N
- The gradient of line C is $\frac{\Delta y}{\Delta x} = \frac{20}{4} = 5 \text{ N}$

A is incorrect because the gradient of this line is $\frac{5}{4} = 1.25 \text{ N}$

B is incorrect because the gradient of this line is $\frac{10}{4} = 2.5 \text{ N}$

D is incorrect because the gradient of this line is $\frac{20 - 10}{4} = 2.5 \text{ N}$

10

The correct answer is **D** because:

- After the collision, the train carriages move together as a single object of mass 15 000 kg and velocity v
- We can find the value of v using the principle of conservation of momentum:
 - The sum of the momentum before the collision equals the sum of the momentum after the collision
- Let's define the positive direction to be to the right
- Momentum before: $(7500 \times 3) - (7500 \times 1.5) = 22\,500 - 11\,250 = 11\,250$
- Momentum after: $(7500 + 7500)v = 15000v$
 - $11\,250 = 15\,000v$
 - $v = \frac{11\,250}{15\,000} = 0.75 \text{ m s}^{-1}$
- Kinetic energy $\left(E_k = \frac{1}{2}mv^2\right)$ before collision = sum of kinetic energies of the train carriages
- Kinetic energy before collision = $\frac{1}{2}(7500)(3)^2 + \frac{1}{2}(7500)(1.5)^2 = 42\,200 \text{ J}$
- Kinetic energy after collision = $\frac{1}{2}(15\,000)(0.75)^2 = 4200 \text{ J}$
 - Kinetic energy lost = $42\,200 - 4200 = 38000 \text{ J}$

